Domain-Aware ESM-2 Based Missense Mutation Pathogenicity Prediction

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Abstract

Missense mutations change a single amino acid in a protein sequence, but their biological effects can range from almost neutral to strongly disease-causing. This makes pathogenicity prediction an important problem in bioinformatics, especially because many variants reported in clinical databases are still uncertain or conflicting. In this project, a lightweight mutation-effect prediction workflow was developed by combining ESM-2 protein language model embedding-change features with biological and domain-aware features collected from ClinVar and UniProt. The workflow used selected human disease genes, generated wild-type and mutant sequence windows around each missense variant, extracted frozen ESM-2 representations, and trained multiple classical machine learning models for benign/pathogenic classification. The proposed ESM + bio/domain RandomForest model achieved 84.49% accuracy, 84.15% pathogenic precision, 95.60% pathogenic recall, 89.51% pathogenic F1-score, and 0.8871 ROC-AUC on the test set. Although the bio/domain-only RandomForest achieved a slightly higher ROC-AUC, the combined model was selected as the final proposed model because it performed better on accuracy, pathogenic recall, F1-score, and the practical selection score. The result suggests that ESM-2 features are most useful when they are combined with interpretable biological context rather than used alone.

1 Introduction

A central challenge in human genomics is deciding whether a missense variant is likely to damage protein function. A missense variant may alter folding, stability, binding, enzymatic activity, localization, or interaction with other molecules. At the same time, many amino-acid substitutions are tolerated. This project focuses on the practical problem of predicting whether a missense mutation is benign-like or pathogenic-like using information that can be collected and processed from ESM-2 protein language model embedding-change features with biological and domain-aware features collected from ClinVar and UniProt. The problem matters because variant interpretation is often a bottleneck in genetic diagnosis and biomedical research. If computational models can prioritize suspicious variants, researchers can focus experimental or clinical follow-up on a smaller group of candidates.

Earlier work on missense variant effect prediction shows that the problem has been approached from several directions, and each direction has its own strengths and weaknesses. Classical tools such as SIFT and PolyPhen-2 use evolutionary conservation, amino-acid substitution patterns, and structural or functional annotations to estimate whether a substitution is likely to be tolerated [1, 2]. They are fast, easy to use, and biologically interpretable, but they can struggle when homologous sequences are limited or when the pathogenic effect depends on context that is not captured by conservation alone. Broader genome-level and ensemble predictors such as CADD and REVEL improved the situation by combining many annotations and existing scores [3, 4]. Their strength is that they perform well across many variant types and genes, but their predictions can be less transparent, and some models may be affected by training-data bias because clinical databases are incomplete and enriched for well-studied genes. Deep generative methods such as EVE showed that evolutionary sequence models can separate disease-causing from benign variants without being trained directly on clinical labels [5]. This is powerful because it reduces label leakage, but these methods still depend heavily on good multiple sequence alignments, which are not equally available for every protein region. Protein language models then became attractive because they learn from raw protein sequences at very large scale. Meier et al. showed that protein language models can make zero-shot mutation-effect predictions from sequence probabilities [6]. ProteinBERT was designed specifically for protein sequences and combined language modeling with Gene Ontology prediction, giving a smaller and faster model that still performed strongly on many protein tasks [7]. Large ESM models further showed that unsupervised protein sequence learning captures structural and functional signals [8, 9]. Brandes et al. used ESM1b for genome-wide missense variant effect prediction and reported strong performance on ClinVar/HGMD and DMS benchmarks, while also showing the advantage of PLMs in regions without MSA coverage and in isoform-specific analysis [10]. However, these PLM approaches can still be computationally heavy, may have sequence-length limitations, and do not directly model splicing or other DNA/RNA-level effects. AlphaMissense achieved very strong proteome-wide pathogenicity prediction by combining AlphaFold-style structure learning and population-frequency information [11], but it is still a large, specialized model and is not as simple to reproduce under resource constraints. More directly related to this project, MutFormer treated reference and mutant proteins as paired sequences and used a transformer with convolutional components for deleterious missense prediction [12]. Its strength is that it explicitly models mutant and wild-type sequence context, but the authors also note limits related to dataset size, computational tuning, and larger-scale validation. Recent papers also show that PLMs can be improved by adding more biological supervision or context. Lafita et al. fine-tuned PLMs using Deep Mutational Scanning data through a Normalised Log-odds Ratio head and evaluated the models on ProteinGym and ClinVar, showing that DMS data can improve variant-effect prediction [13]. P13LG and ProMEP moved toward multimodal sequence-plus-structure learning, where sequence representations are combined with protein geometry to improve blind mutation-effect prediction and protein engineering [14, 15]. These methods are powerful, especially for structural and epistatic effects, but they require structural inputs or substantially more computational power. VespaG used ESM-2 embeddings and a GEMME-guided training strategy to produce very fast fitness predictions, which is useful because speed and accessibility matter for large mutational landscapes [16]. Saadat and Fellay fine-tuned ESM2 and ProtT5 to predict amino-acid-level protein features and then used reference-versus-mutant feature changes to interpret missense variants, which is especially relevant to this project because it connects PLM predictions with functional-region interpretation [17]. Finally, hint token learning was proposed to make mutation positions more explicit to PLMs; it improved several mutational-function prediction tasks and suggested that directing model attention toward functional domains can make predictions more biologically meaningful [18]. Overall, the literature shows a clear trend: the best modern approaches use either evolutionary information, protein language models, structural context, experimental DMS data, or some combination of these. The gap that remains is the need for a lightweight, reproducible workflow that can run on lower computational resources, avoid full PLM fine-tuning, and still combine ESM-derived mutation embedding changes with interpretable biochemical and UniProt domain-aware features.

Although recent studies have shown that protein language models, structure-aware models, and fine-tuned deep learning approaches can achieve strong performance in mutation effect prediction, many of these methods are computationally expensive, difficult to reproduce within a limited academic setting, or dependent on large-scale training data, multiple sequence alignments, experimentally measured DMS scores, or structural information. There is therefore a practical need for a lightweight but biologically meaningful approach that can still capture useful mutation-level signals without requiring high-end computational resources. Our work addresses this gap by combining frozen ESM2-derived embedding-change features with simple but interpretable biological and domain-aware descriptors such as BLOSUM62 score, hydropathy change, mass change, normalized mutation position, and UniProt feature annotations. Instead of training a large protein language model from scratch or fine-tuning millions of parameters, the proposed method uses ESM2 only as a feature extractor and applies classical machine-learning models for final classification. This makes the workflow easier to run, understand, and reproduce under computational resource constraints while still remaining connected to recent advances in protein language model-based variant prediction. More importantly, the domain-aware design improves interpretability, because the prediction is not only based on hidden neural embeddings but also on biologically explainable features related to amino-acid substitution severity and functional protein regions. Therefore, this project is important because it presents a practical, resource-efficient, and interpretable framework for prioritizing pathogenic missense variants and variants of uncertain significance.

1.1 Our Contribution

- Collected a ClinVar-based missense variant dataset from selected human disease genes.
- Mapped variants to canonical UniProt protein sequences and removed entries with coordinate or amino-acid mismatches.
- Generated local wild-type and mutant sequence windows around each mutation to keep ESM-2 inference feasible in free Colab GPU.
- Extracted ESM-2 embedding-change features comparing wild-type and mutant windows at both window level and mutation-site level.
- Engineered interpretable biological features including BLOSUM62 score, hydropathy change, mass change, charge change, proline/glycine indicators, normalized position, and UniProt domain/region indicators.
- Compared ESM-only, bio/domain-only, and combined ESM + bio/domain feature sets using Logistic Regression and RandomForest classifiers.
- Performed an ablation-style comparison and a VUS/conflicting-variant case study to show how the final model can prioritize uncertain variants.

2 Material and Methods

Figure 1 shows the complete pipeline used in this project. First, missense variants and protein-related annotations were collected from ClinVar and UniProt. The data were then cleaned and prepared by filtering labeled variants, defining benign/pathogenic classes, and creating the required train/test and VUS sets. Next, two feature groups were extracted: ESM2-based embedding-change features from local wild-type and mutant sequence windows, and interpretable biological/domain-aware features such as substitution score, physicochemical changes, normalized position, and UniProt functional-region indicators. These feature groups were fused into a final representation for each variant, after which multiple classical machine-learning models were trained and compared. Finally, the best-performing model was selected based on test-set evaluation metrics and used as the final proposed framework for variant prioritization.

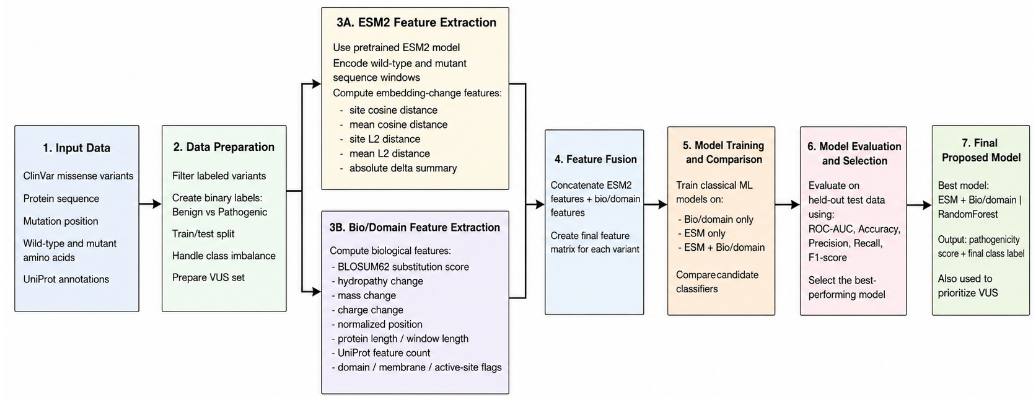

Figure 1: Overall workflow of the proposed missense pathogenicity prediction pipeline.

2.1 Dataset Description

ClinVar was used to collect missense variants and clinical significance labels [19]. From the selected nine disease-associated genes, 23,552 target missense variants were initially parsed. After label filtering, sampling, and sequence verification, 1,441 labeled variants were retained for supervised training and testing, while 60 VUS/conflicting variants were kept separately for the case study. UniProt was used to collect canonical protein sequences and protein-level annotations for the same nine genes, giving 9 protein records and 3,765 total feature annotations [20].

Dataset component	**Value**
Labeled training/test variants	1,441
Benign / likely benign	445
Pathogenic / likely pathogenic	996
Genes included	9
VUS/conflicting variants for case study	60
Main data sources	ClinVar, UniProt

Table 1: Summary of the dataset used in the project.

The final labeled dataset contained 1,441 missense variants from nine human disease-related genes: BRCA1, BRCA2, CFTR, G6PD, HBB, LDLR, PAH, PTEN, and TP53. Only variants labelled benign/likely benign or pathogenic/likely pathogenic were used for supervised training. In this dataset, 445 variants were benign or likely benign and 996 variants were pathogenic or likely pathogenic. Variants of uncertain significance or conflicting clinical significance were not used for training; they were saved for the later case study. This separation is important because uncertain variants should not be treated as ground truth labels.

GeneSymbol	Benign	Pathogenic	Total
BRCA1	120	120	240
BRCA2	120	105	225
CFTR	15	120	135
G6PD	4	90	94
HBB	18	81	99
LDLR	40	120	160
PAH	1	120	121
PTEN	7	120	127
TP53	120	120	240

Table 2: Gene-wise distribution of benign and pathogenic variants.

Feature Group	Examples	Purpose
ESM embedding-change features	esm_site_cosine_distance, esm_mean_l2_distance, esm_site_abs_delta_mean	Measure how much the mutation changes learned protein-language-model representation
Biochemical substitution features	BLOSUM62, hydropathy delta, mass delta, charge change, to_proline	Describe chemical and evolutionary severity of the amino-acid replacement
Position/protein context features	protein_length, window_length, normalized mutation position	Capture where the mutation occurs in the protein sequence
UniProt domain-aware features	in_domain_like_feature, in_active_or_binding_feature, uniprot_feature_count	Capture whether the mutation falls in a biologically annotated region

Table 3: Main feature groups used in the proposed model.

2.2 Tools / Models / Algorithms Used

The central representation model was ESM-2 35M, used only for inference and feature extraction. This was chosen because it is much lighter than very large protein language models and can be handled in free Colab GPU when local sequence windows are used. The model was not fine-tuned; this kept the computation realistic and reduced overfitting risk. The classification models were Logistic Regression and RandomForest. Logistic Regression was used as a simple linear baseline, while RandomForest was used because it can capture nonlinear relationships among mutation position, biochemical changes, domain annotations, and ESM distances.

Model setup	Input features	Purpose
ESM-only + Logistic Regression	6 ESM features	Linear baseline for language-model features
ESM-only + RandomForest	6 ESM features	Nonlinear baseline for language-model features
Bio/domain-only + Logistic Regression	17 bio/domain features	Linear baseline for interpretable biological features
Bio/domain-only + RandomForest	17 bio/domain features	Strong baseline without ESM features
ESM + bio/domain + Logistic Regression	23 combined features	Combined linear model
ESM + bio/domain + RandomForest	23 combined features	Final proposed nonlinear combined model

Table 4: Models compared in the project.

After feature extraction, two classical machine-learning models were used: Logistic Regression and RandomForest. Both models were trained on three feature settings: **bio/domain only**, **ESM only**, and the proposed **ESM + bio/domain** feature set. This was done to compare whether combining protein language model features with biological/domain-aware features improves prediction performance.

For **Logistic Regression**, the main settings were `max_iter`=3000, `class_weight`="balanced", and `random_state`=42. The value `max_iter`=3000 was used to allow enough iterations for convergence. `class_weight`="balanced" was used because the dataset had more pathogenic variants than benign variants, so weighting helped reduce bias toward the majority pathogenic class. `random_state`=42 was used for reproducibility. A small sensitivity test was also performed using different values of the regularization parameter `C`: 0.01, 0.1, 1.0, and 10.0. The `C` value controls regularization strength, where smaller values apply stronger regularization. The combined ESM + bio/domain feature set performed best for Logistic Regression, especially around `C`=0.1 and `C`=1.0.

For **RandomForest**, the final settings were `n_estimators`=300, `max_depth`=None, `min_samples_leaf`=2, `class_weight`="balanced", `random_state`=42, and `n_jobs`=-1. The number of trees was tested using `n_estimators`=100, 200, 300, and 500. The proposed ESM + bio/domain RandomForest achieved its best validation performance at `n_estimators`=300, so 300 trees were selected for the final model. `max_depth`=None allowed the trees to learn complex non-linear patterns, while `min_samples_leaf`=2 helped reduce overfitting by preventing overly specific splits. `n_jobs`=-1 was used to speed up training by using all available CPU cores in Colab.

Overall, Logistic Regression served as a simple linear baseline, while RandomForest was chosen as the final classifier because it performed better and could capture non-linear relationships among ESM embedding-change features, amino-acid substitution features, and UniProt domain-aware features. The final proposed model was therefore **ESM + bio/domain RandomForest** with 300 trees.

2.3 Performance Evaluation

The dataset was divided into training and test sets using a stratified 75/25 split with `random_state = 42`. Stratification was used so that the benign/pathogenic ratio remained similar in both sets. The main metrics were accuracy, pathogenic precision, pathogenic recall, pathogenic F1-score, and ROC-AUC. For final model selection, a practical screening score was calculated as the mean of accuracy, pathogenic recall, and pathogenic F1-score. This choice was made because the project goal is not only to rank variants but also to correctly flag pathogenic-like variants at a useful decision threshold. In this setting, missing a truly pathogenic variant is usually more serious than flagging some extra benign variants for follow-up.

The metrics were interpreted as follows: accuracy measures overall correctness; precision measures how many predicted pathogenic variants are actually pathogenic; recall measures how many true pathogenic variants are caught; F1-score balances precision and recall; ROC-AUC measures ranking quality across possible probability thresholds.

3 Experimental Analysis

The link to the Google Colab Notebook is given here.

3.1 Model Comparison and Ablation Study

Table 5 shows the main model comparison. This table is also the ablation study because it compares three feature settings: ESM-only, bio/domain-only, and ESM + bio/domain. The ESM-only models were weaker than the combined models. The bio/domain-only RandomForest was a strong baseline and achieved the highest ROC-AUC. However, the combined ESM + bio/domain RandomForest achieved the highest accuracy, pathogenic recall, F1-score, and selection score. This is the main evidence that the ESM features add complementary information when they are combined with biological context.

The proposed model reached a selection score of 0.8987, compared with 0.8795 for the bio/domain-only RandomForest. The improvement is not huge, but it is meaningful for this project because it occurs in the metrics that matter most for practical screening: accuracy, recall, and F1-score. The bio/domain-only RandomForest still had a slightly higher ROC-AUC, so it can be concluded that ESM features did not dominate the task alone, but they improved the final combined screening model.

3.2 Hyperparameter Sensitivity Analysis

To satisfy the experimental analysis requirement, a small hyperparameter sensitivity test was performed for the two main classical machine-learning algorithms used in this project: Logistic Regression and RandomForest. For Logistic Regression, the regularization parameter `C` was varied using four values: 0.01, 0.1,

Model	**Features**	**Accuracy**	**Precision**	**Recall**	**F1**	**ROC-AUC**	**Selection score**
ESM + bio/domain RandomForest	23	0.8449	0.8415	0.9560	0.8951	0.8871	0.8987
Bio/domain only RandomForest	17	0.8310	0.8462	0.9240	0.8834	0.9000	0.8795
ESM only RandomForest	6	0.7452	0.7724	0.8960	0.8296	0.7446	0.8236
ESM + bio/domain Logistic Regression	23	0.7147	0.8356	0.7320	0.7804	0.7908	0.7424
Bio/domain only Logistic Regression	17	0.6787	0.8190	0.6880	0.7478	0.7409	0.7048
ESM only Logistic Regression	6	0.6870	0.8408	0.6760	0.7494	0.7526	0.7041

Table 5: Main model comparison and ablation study. Selection score = mean(accuracy, recall, F1).

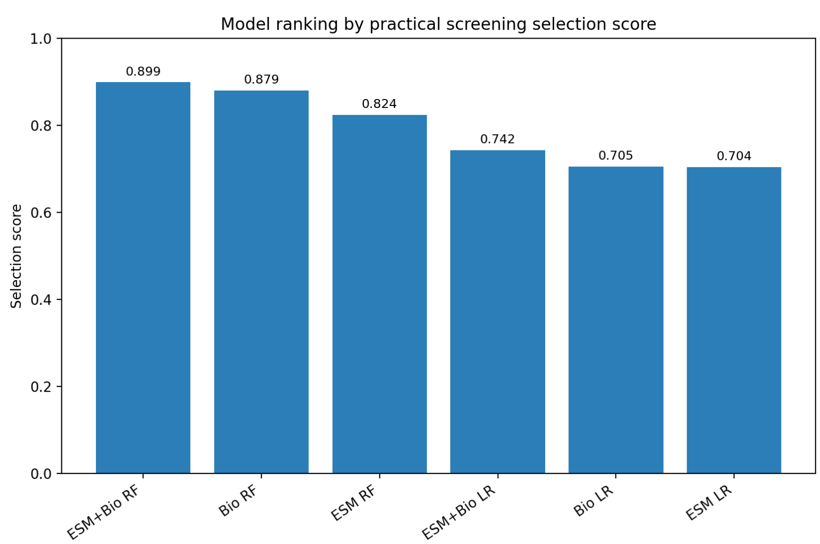

Figure 2: Model ranking by practical screening selection score.

1.0, and 10.0. For RandomForest, the number of trees, `n_estimators`, was varied using 100, 200, 300, and 500 trees. This test was performed on three feature settings: bio/domain features only, ESM features only, and the combined ESM + bio/domain feature set. The models were evaluated using validation F1-score for the pathogenic class, along with accuracy, precision, recall, ROC-AUC, and a practical selection score.

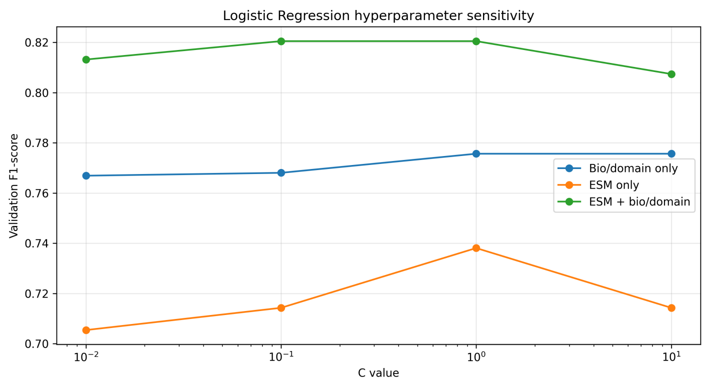

Figure 3: Hyperparameter sensitivity analysis for Logistic Regression across the evaluated feature settings.

The Logistic Regression sensitivity plot shows that the combined ESM + bio/domain feature set consistently outperformed the ESM-only and bio/domain-only feature sets across most `C` values. The best validation F1-score for Logistic Regression was obtained by the ESM + bio/domain model at `C` = 0.1 and `C` = 1.0, both giving an F1-score of approximately **0.821**. In comparison, the best bio/domain-only Logistic Regression model achieved an F1-score of approximately **0.776**, while the best ESM-only Logistic Regression model achieved an F1-score of approximately **0.738**. This indicates that the combined feature set provides more useful information for a linear classifier than either feature group alone. However, Logistic Regression still performed lower than RandomForest overall, suggesting that the relationship between the features and pathogenicity may not be fully linear.

The RandomForest sensitivity plot shows that RandomForest was relatively stable across different numbers of trees. For the proposed ESM + bio/domain feature set, the best validation F1-score was obtained at `n_estimators` = 300, with an F1-score of approximately **0.912**. The bio/domain-only RandomForest model also achieved a similar F1-score of approximately **0.912** at 100, 200, and 500 trees, while the ESM-only RandomForest model remained clearly lower, with its best F1-score around **0.821** at 500 trees. This result shows that ESM features alone were not sufficient for the strongest prediction, but when combined with biological and domain-aware features, they helped build a strong final model. The small differences between 100, 200, 300, and 500 trees also indicate that the RandomForest model was not highly sensitive to the exact number of trees.

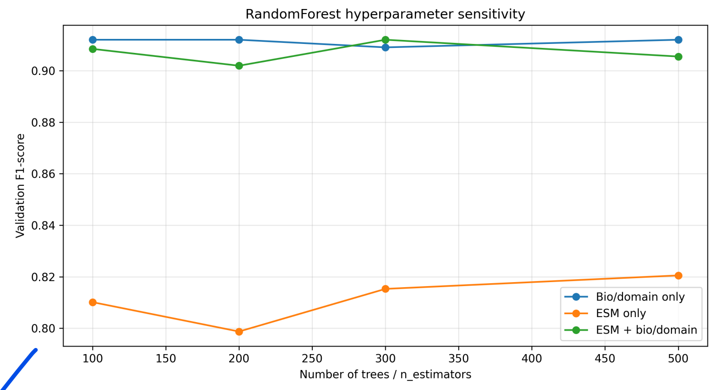

Figure 4: Hyperparameter sensitivity analysis for RandomForest across the evaluated feature settings.

Algorithm	**Feature set**	**Best hyperparameter**	**Validation F1-score**	**ROC-AUC**	**Selection score**
Logistic Regression	Bio/domain only	`C = 1.0 or 10.0`	0.776	0.782	0.729
Logistic Regression	ESM only	`C = 1.0`	0.738	0.817	0.686
Logistic Regression	ESM + bio/domain	`C = 0.1 or 1.0`	0.821	0.828–0.836	0.782
RandomForest	Bio/domain only	`n_estimators = 100/200/500`	0.912	0.900–0.903	0.909
RandomForest	ESM only	`n_estimators = 500`	0.821	0.761	0.807
RandomForest	ESM + bio/domain	`n_estimators = 300`	0.912	0.913	0.909

Table 6: Best hyperparameter settings and validation performance for Logistic Regression and RandomForest across the evaluated feature sets.

Overall, the hyperparameter sensitivity analysis supports the final choice of **ESM + bio/domain RandomForest** as the proposed model. Although the bio/domain-only RandomForest achieved a very similar validation F1-score during this validation experiment, the proposed combined model achieved the strongest balance between validation F1-score, ROC-AUC, and practical selection score while also preserving the central project idea of combining protein language model features with interpretable biological features.

3.3 Learning Curve and Stability Check

Because Logistic Regression and RandomForest are classical machine-learning models, they do not produce epoch-wise convergence curves like deep neural networks. Therefore, learning curves were used as a suitable alternative convergence and stability check. In these curves, the models were trained using increasing numbers of training examples, and both training F1-score and cross-validation F1-score were plotted. This shows whether the models improve with more training data and whether there is a large gap between training and validation performance.

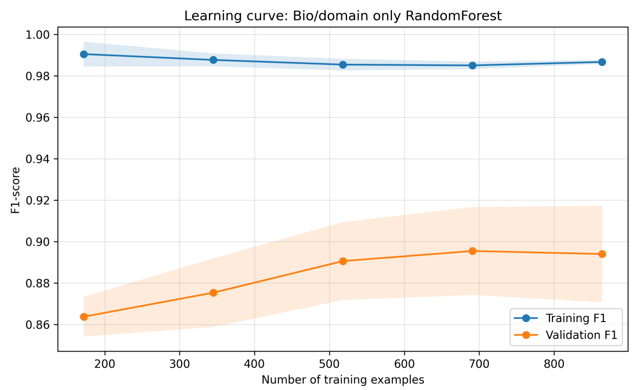

Figure 5: Learning curve for the bio/domain RandomForest model.

The bio/domain-only RandomForest learning curve shows that the training F1-score stayed very high, close to **0.99**, across all training sizes. The validation F1-score increased from approximately **0.864** at 172 training examples to approximately **0.894–0.896** at larger training sizes. This indicates that the model benefits from additional training data and becomes more stable as more examples are used. However, the noticeable gap between training and validation F1-score suggests mild overfitting, which is expected for RandomForest models trained on relatively small biological datasets.

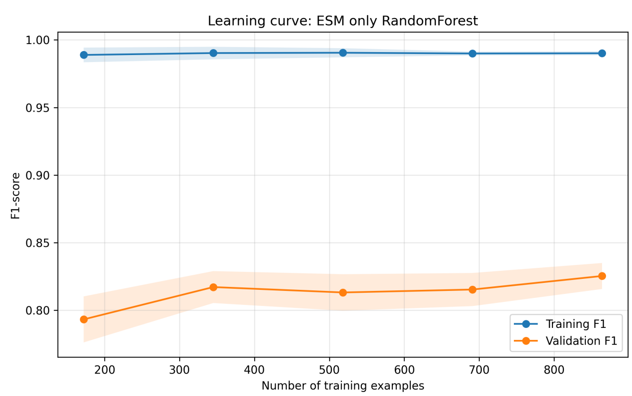

Figure 6: Learning curve for the ESM-only RandomForest model.

The ESM-only RandomForest learning curve shows weaker validation performance compared with the other two RandomForest models. The training F1-score remained very high, close to **0.99**, but the validation F1-score stayed lower, rising only from approximately **0.793** to around **0.825** as the number of training examples increased. This result suggests that ESM-derived embedding-change features alone contain useful mutation information, but they are not enough to achieve the best classification performance in this project. The model still overfits the training data and generalizes less strongly than models using biological/domain-aware features.

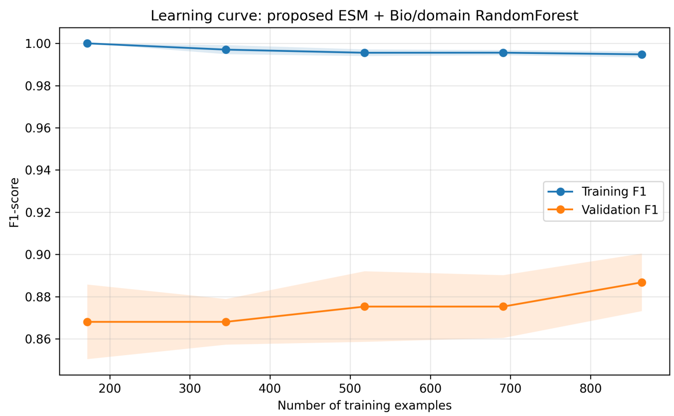

Figure 7: Learning curve for the proposed ESM + bio/domain RandomForest model.

The proposed ESM + bio/domain RandomForest learning curve shows that the training F1-score remained very high, decreasing slightly from **1.000** to approximately **0.995** as the training size increased. The validation F1-score improved from approximately **0.868** at 172 training examples to approximately **0.887** at 864 training examples. This upward trend indicates that the proposed model gains generalization benefit from more training data. Although there is still a gap between the training and validation curves, the validation curve improves gradually and remains stable, showing that the model is learning useful patterns rather than behaving randomly.

Training examples	**Training F1 mean**	**Validation F1 mean**
172	1.000	0.868
345	0.997	0.868
518	0.996	0.875
691	0.996	0.875
864	0.995	0.887

Table 7: Learning-curve summary for the proposed ESM + bio/domain RandomForest model.

Overall, the learning curves show that all RandomForest models achieved very high training performance, but their validation performance differed. The ESM-only model had the lowest validation F1-score, showing that ESM features alone were not sufficient. The bio/domain-only and ESM + bio/domain models performed better, confirming that biological and domain-aware features are important for missense variant classification. The proposed ESM + bio/domain RandomForest model showed stable improvement in validation F1-score with increasing training data, supporting its use as the final model. However, the gap between training and validation F1-score also suggests that future work should include larger datasets, stronger regularization, and external gene-level validation to further reduce overfitting.

3.4 Detailed Result of the Proposed Model

The final proposed model is the ESM + bio/domain RandomForest. Its confusion matrix is especially useful for understanding its behavior. Out of 250 pathogenic variants in the test set, it correctly identified 239 and missed only 11. This gives a pathogenic recall of 95.60%. On the other hand, it misclassified 45 benign variants as pathogenic-like, which means the model tends to be aggressive. In a screening context, this behavior is acceptable because false positives can be checked later, while false negatives are more dangerous.

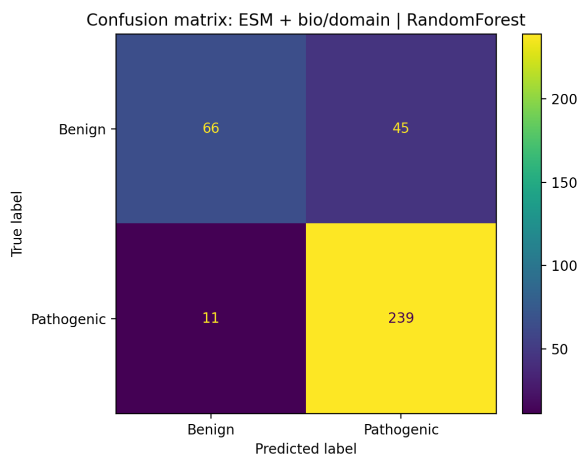

Figure 8: Confusion matrix of the proposed ESM + bio/domain RandomForest model.

The ROC curve in Figure 9 gives an AUC of about 0.887, which indicates good ranking ability. The value is slightly below the bio/domain-only RandomForest ROC-AUC of 0.900, but the proposed model performed better at the selected operating point. This is why the proposed model is described as the best practical classifier rather than the absolute best model under every metric.

3.5 Feature Importance and Biological Interpretation

Feature importance is one of the strongest parts of this project because it makes the model more interpretable. The top features show that the RandomForest used a mixture of protein-context information, ESM-derived mutation-change

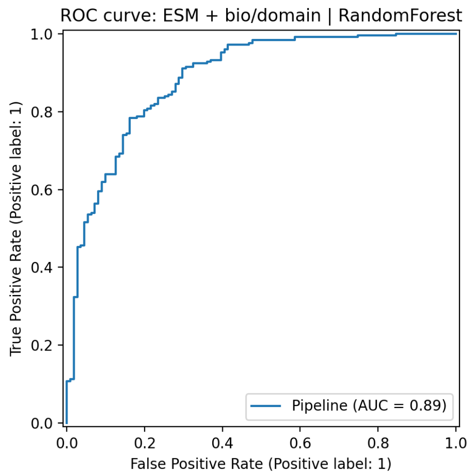

Figure 9: ROC curve of the proposed ESM + bio/domain RandomForest model.

features, and biochemical domain features. The most important feature was `protein_length`, followed by `window_length` and normalized mutation position. This suggests that the model is learning gene/protein context. At the same time, several ESM features appear among the top-ranked predictors, including `esm_site_cosine_distance`, `esm_mean_cosine_distance`, `esm_site_abs_delta_mean`, `esm_mean_l2_distance`, and `esm_site_l2_distance`. This supports the central claim that ESM embedding changes contributed to the final model.

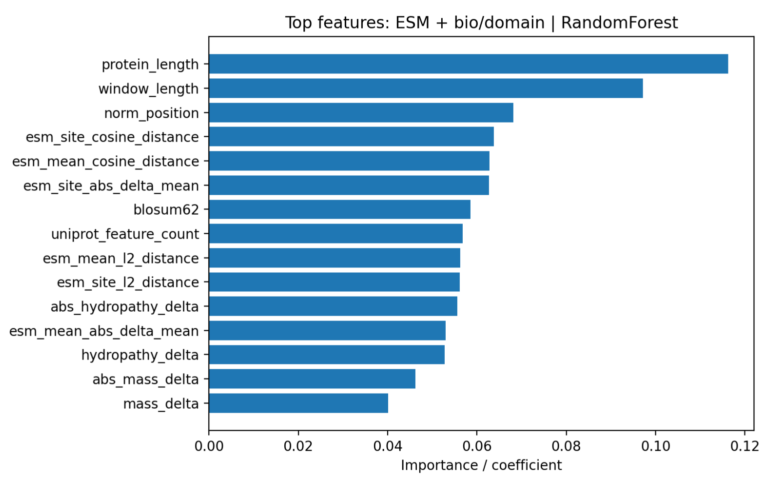

Figure 10: Top-ranked features used by the proposed ESM + bio/domain RandomForest model.

Feature	RandomForest importance
protein_length	0.1163
window_length	0.0971
norm_position	0.0681
esm_site_cosine_distance	0.0638
esm_mean_cosine_distance	0.0628
esm_site_abs_delta_mean	0.0627
blosum62	0.0586
uniprot_feature_count	0.0568
esm_mean_l2_distance	0.0563
esm_site_l2_distance	0.0561
abs_hydropathy_delta	0.0556
esm_mean_abs_delta_mean	0.0529

Table 8: Top 12 features in the proposed RandomForest model.

The biological features also behaved in an expected way. Pathogenic variants had a lower average BLOSUM62 score, larger hydropathy change, larger mass change, and more frequent charge changes than benign variants. This means the model is not only finding statistical patterns but is also using features that make biological sense.

3.6 VUS and Conflict-Variant Case Study

After training, the final model was applied to 60 variants of uncertain significance or conflicting clinical significance. These variants were not used as training labels. The goal was not to clinically classify them, but to show how the model could prioritize variants for further investigation. The model predicted 16 variants as pathogenic-like and 44 as benign-like. The top pathogenic-like predictions from the latest uploaded VUS file are shown in Table 7.

For example, PTEN Leu100Pro received the highest predicted pathogenic probability in this case study. This is biologically plausible as a candidate for follow-up because the mutation introduces proline inside a domain-like region. However, the result should still be interpreted as computational prioritization only. A model like this cannot replace experimental validation, expert clinical review, or established variant-interpretation guidelines.

3.7 Limitations of the Experiment

- The dataset is imbalanced, with more pathogenic than benign variants. Class weighting helps, but it does not fully remove the effect of the imbalance.
- Some genes are strongly skewed toward pathogenic examples. For example, PAH, PTEN, G6PD, and CFTR have very few benign variants in this selected dataset. This can cause gene-level bias.

Gene	Variant	Pathogenic probability	Predicted class	Domain-like?	Covering UniProt feature types
PTEN	NM_000314.8(PTEN):c.299T>C (p.Leu100Pro)	0.9657	Predicted pathogenic-like	1	Chain;Domain;Helix; Natural variant
PAH	NM_000277.3(PAH):c.1194A>C (p.Lys398Asn)	0.9210	Predicted pathogenic-like	0	Chain;Helix
LDLR	NM_000527.5(LDLR):c.1474G> T(p.Asp492Tyr)	0.8847	Predicted pathogenic-like	1	Beta strand;Chain; Natural variant;Repeat; Topological domain
CFTR	NM_000492.4(CFTR):c.3689T> G(p.Ile1230Ser)	0.8631	Predicted pathogenic-like	1	Alternative sequence; Beta strand;Chain; Domain;Natural variant;Topological domain
CFTR	NM_000492.4(CFTR):c.89A>C (p.Gln30Pro)	0.8127	Predicted pathogenic-like	0	Chain;Topological domain
CFTR	NM_000492.4(CFTR):c.2755T> A(p.Tyr919Asn)	0.7988	Predicted pathogenic-like	1	Alternative sequence; Chain;Domain;Helix; Natural variant; Transmembrane
G6PD	NM_001360016.2(G6PD):c.937G >A(p.Asp313Asn)	0.7310	Predicted pathogenic-like	0	Beta strand;Chain
BRCA2	NM_000059.4(BRCA2):c.193C> T(p.Pro65Ser)	0.7304	Predicted pathogenic-like	1	Chain;Region
BRCA1	NM_007294.4(BRCA1):c.5085T> A(p.Phe1695Leu)	0.6622	Predicted pathogenic-like	1	Alternative sequence; Beta strand;Chain; Domain;Natural variant
BRCA2	NM_000059.4(BRCA2):c.7844T> A(p.Ile2615Asn)	0.6538	Predicted pathogenic-like	1	Chain;Region

Table 9: Top VUS/conflicting variants prioritized by the proposed model. These are not clinical diagnoses.

- `protein_length` and `window_length` are highly important features. Because each gene has a characteristic protein length, the model may partly learn gene identity rather than pure mutation effect.
- The test set was created from the same gene panel used during training. A stricter validation would test the model on genes that were completely unseen during training.
- Only local sequence windows were used. This keeps the project lightweight, but it may miss long-range effects involving distant residues or domains.
- The model predicts pathogenic-like probability, not clinical truth. VUS predictions should be treated only as prioritization suggestions.

4 Conclusion

This project developed a lightweight domain-aware missense mutation predictor using ClinVar labels, UniProt protein annotations, ESM-2 embedding-change features, and classical machine learning classifiers. The best practical model was the ESM + bio/domain RandomForest, which achieved 84.49% accuracy, 95.60% pathogenic recall, 89.51% pathogenic F1-score, and 0.887 ROC-AUC. The model was selected because it performed best on the practical screening score based on accuracy, recall, and F1-score.

The most important conclusion is that ESM-2 features alone were not enough to outperform the biological feature baseline. However, when ESM features were combined with domain-aware and biochemical features, they helped produce the strongest practical classifier. This makes the project result realistic and valuable: protein language models are powerful, but for small applied bioinformatics projects, they work best when paired with interpretable biological information.

Future work should include a larger and more balanced dataset, leave-one-gene-out validation, external testing on unseen genes, comparison with established tools such as SIFT, PolyPhen-2, CADD, REVEL, EVE, and AlphaMissense, and possible addition of structure-based features from AlphaFold models. A future version could also calibrate the prediction threshold depending on whether the user wants higher recall or higher precision.

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